

Merged

One chain, two decisions

The combined physics, parameters and objective for the Baghewala digital twin — source for the deck, and for model training.

PROBLEM STATEMENT

SIH26120 · Oil India Limited · Software · Smart Automation

USE FOR

Presentation content · ML parameter ranges · Judge Q&A

OPEN ITEMS

Three, listed in section six

One

The problem

Baghewala, Rajasthan. Heavy crude at 14–19° API, roughly 11,500 cP at the native reservoir temperature of 46–50 °C. At that thickness it will not flow, and reservoir pressure is too low to push it anyway. Two interventions are mandatory: steam to thin the oil, and a sucker rod pump to lift it.

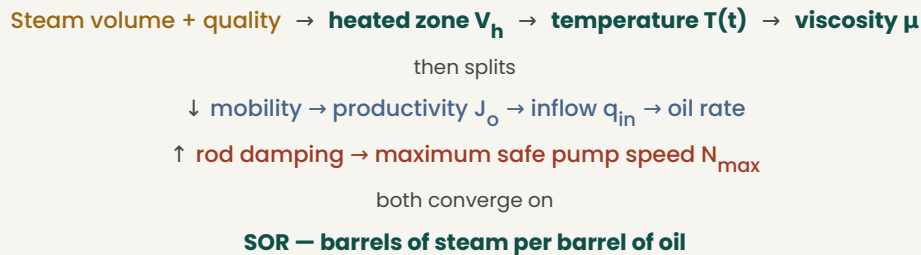
Steam heats the near-wellbore rock. Over the following weeks that heat drains into the surrounding cold formation, viscosity climbs back, and two things degrade at once — less oil reaches the well, and the rod string can no longer fall through the thickening oil on its downstroke. It goes slack, gets shock-loaded on every stroke, and fatigues until it fractures.

Both effects have one cause: reservoir temperature. Steam scheduling and pump operation are currently managed by separate teams on historical practice, so nothing connects them.

That disconnect is the problem statement. The deliverable is a well-to-surface digital twin that observes the current state, predicts how it evolves, and prescribes both decisions from the same prediction.

Two

The causal chain



The split is the key structural point. Production potential and mechanical safety are not two problems that happen to share a well. They are two consequences of one variable, which is why optimising them separately is guaranteed to be suboptimal.

Heated zone volume matters beyond its obvious role: a larger heated zone has proportionally less surface exposed to cold rock, so it cools more slowly. Steam volume therefore buys *duration*, not just temperature. This is the physical variable through which "inject more now" versus "wait" becomes computable rather than habitual.

Three

The objective

Maximise oil produced per unit of steam injected, subject to pump speed staying below the rod-floating limit.

Stated formally, over a cycle:

OBJECTIVE FUNCTION

$$\min \text{ SOR} = V_{\text{steam}} / N_p \quad \text{subject to} \quad N(t) \leq N_{\text{max}}(\mu(t))$$

Decision variables: steam volume V_s , soak time t_{soak} , re-steam date t_{cut} , and pump speed $N(t)$ each day.

N_p = cumulative oil over the cycle, obtained by integrating $q_{\text{in}} \times \text{fillage}$.

Two consequences worth stating explicitly in the deck.

Rod safety is a constraint, not the goal. If avoiding failures were the objective, the optimum would be to run at minimum speed forever — no failures, almost no oil. Failures matter because they halt production and incur workover cost. The value of knowing N_{max} precisely is that it lets the pump run *close* to the limit rather than conservatively far below it.

Re-steam timing moves SOR more than pump speed does. Pump speed is a daily optimisation within a cycle. Re-steam timing is the cycle-level decision, and it directly sets the numerator of SOR. Too early wastes steam on a well still producing; too late means weeks of crawling output at 4 SPM.

Four Equations

1 · STEAM → HEATED ZONE [ASSUMPTION — reduced-order energy balance]

$$V_h = \eta \cdot m_s \cdot [x \cdot L + C_p(T_s - T_i)] / [M_R(T_s - T_i)]$$

Steam enthalpy divided by rock heat requirement gives heated rock volume. Radius $r_h = \sqrt{(V_h / \pi h)}$.

Worked: 2,000 m³ CWE, $x = 0.70$, $T_s = 250$ °C, $\eta = 0.5 \rightarrow 2039$ kJ/kg $\rightarrow V_h = 4,339$ m³, $r_h = 11.8$ m

Feeds: the effective heat capacity $C_{\text{eff}} = M_{\text{eff}} c_{p,\text{eff}}$ in equation 2, where $M_{\text{eff}} = \rho \cdot V_h$. Without this, C_{eff} is an uncalibrated fudge factor.

2 · THERMAL BALANCE → TEMPERATURE [PHYSICS]

$$C_{\text{eff}} \cdot dT_r/dt = Q_{\text{steam}} - Q_{\text{loss}} - Q_{\text{prod}}$$

General form, valid during injection, soak and production. For the production phase with steam off and loss proportional to temperature gap, this integrates to a closed form usable directly in a dashboard:

T(t) = $T_i + (T_s - T_i) \cdot e^{-t/\tau}$ where $\tau = C_{\text{eff}}/U_{\text{eff}}$ and τ grows with r_h .

Worked ($\tau = 30$ d): day 0 $\rightarrow 250$ °C · day 30 $\rightarrow 124$ °C · day 60 $\rightarrow 77$ °C · day 90 $\rightarrow 60$ °C

3 · TEMPERATURE → VISCOSITY [PHYSICS — ASTM D341 / Walther, one constant calibrated]

$$\log_{10}(\log_{10}(v + 0.7)) = A - B \cdot \log_{10}(T)$$

T in kelvin, v in cSt, μ (cP) = $v \times \text{SG}$. Anchored on the measured 11,500 cP at 50 °C with $B = 3.5 \rightarrow A = 9.394$.

Worked: day 0 $\rightarrow 5$ cP · day 30 $\rightarrow 93$ cP · day 60 $\rightarrow 1,148$ cP · day 90 $\rightarrow 4,465$ cP · fully cold $\rightarrow 11,500$ cP

Do not substitute a simple exponential. Fitting $\mu = A \cdot e^{B/T}$ to oil produces errors above 700% at mid-range temperatures. The double logarithm exists because oil viscosity varies too violently for an ordinary curve.

4 · VISCOSITY → PRODUCTION [PHYSICS]

$$J_o \propto k \cdot k_{ro} / \mu_o \quad \cdot \quad q_{in} = J_o (P_r - P_{wf})$$

Thin oil flows more readily, so the productivity index rises as viscosity falls. Inflow is productivity times drawdown.

Coupled to the pump by: $P_{intake} = P_{wh} + \Delta P_h - \Delta P_f - \Delta P_{other}$, with $P_{wf} \approx P_{intake}$.

Surface rate: $Q_{theoretical} = A_p \cdot S_{eff} \cdot N$, actual = theoretical $\times$ fillage F .

Limitation: linear productivity index assumed; non-linear Vogel-type inflow at high drawdown is out of scope for vl.

5 · VISCOSITY → PUMP SPEED LIMIT [ASSUMPTION — static force balance, ours, needs calibration]

$$N_{max} = 30 \cdot v_f / S \quad \text{where} \quad v_f = (\rho_s - \rho_o) \cdot g \cdot d_r^2 \cdot \ln(d_t / d_r) / (8\mu)$$

Rod weight against viscous drag gives the fall speed available; the pump allows a half-stroke period of $30/N$ seconds to complete it. If required exceeds available, the string goes slack.

Worked ($S = 2.5$ m): day 0–60 → no binding limit · day 90 → **11.5 SPM** · fully cold → **4.5 SPM**

Key finding for the deck: the limit does not bite for roughly the first two months of a cycle, then collapses rapidly. That narrow end-of-cycle window is where the mechanical damage occurs, and it is precisely the period nobody currently monitors.

6 · ROD DYNAMICS AND CARD DIAGNOSTICS [PHYSICS — lumped, vl]

$$M_{eff} \cdot \ddot{x} + C_{eff} \cdot \dot{x} + K_{eff} \cdot x = F_{ext}(t)$$

Generates a predicted dynamometer card. Comparing predicted against observed gives a residual $R_{card} = D_{observed} - D_{predicted}$ used to classify underfill, valve faults, leakage and mechanical anomaly.

Note: equation 5 answers the floating question analytically without integrating this, so floating detection need not depend on card fidelity. Reserve this model for fault classification, where card *shape* genuinely carries the signal.

Where ML attaches: Corrected = Physics + ML_residual, applied at the wellbore and dynamometer modules. Physics stays the backbone; models learn the discrepancy rather than replacing the mechanism. Two models fall out — one predicting the temperature decay curve for this well and cycle, one solving the constrained optimisation in section three.

Five

Parameters for model training

Confidence tiers — **A** stated in the problem statement, **B** published Oil India or SPE data, **D** our assumption, requires calibration.

Symbol	Meaning	Unit	Value / range	Conf.
°API	Crude gravity	°API	17–19 (PS) vs 14–17 (field)	A / B
T_i	Native reservoir temperature	°C	46–48 (PS) vs ~50 BHT (field)	A / B
μ_o	Viscosity at T _i , at 50 s ⁻¹	cP	10,000–13,000	B
D	Well depth, Jodhpur Sandstone	m	~1150	B
P_r	Reservoir pressure	ksc / MPa	~116 ksc = 11.4 MPa	B

ϕ	Porosity	fraction	< 0.10	B
V_s	Steam volume per cycle	m ³ CWE	300 – 12,000	B
x	Steam quality at wellhead	fraction	0.60 – 0.80	B
T_s	Steam temperature	°C	200 – 300	B
P_{inj}	Injection pressure	MPa	1 – 11 (≤ 1600 psig preferred)	B
t_{soak}	Soak time	days	3 – 14	B
t_{cut}	Cycle length before re-steam	days	60 – 365	B
N	Pump speed	SPM	2 – 12	D
S	Stroke length	m	1.0 – 3.0	D
SOR	Steam–oil ratio	—	3.0 – 5.2	B
RF	Recovery factor	%	<20 typical, up to 35	B
q_o	Field production rate	bbl/day	~655 (July 2025)	B
M_R	Rock volumetric heat capacity	J/(m ³ ·K)	2.35×10^6	D
η	Thermal efficiency	fraction	0.4 – 0.7	D
τ	Cooling time constant	days	15 – 50	D
A, B	Walther constants	—	$A = 9.394, B = 3.5$	D
ρ_o / ρ_s	Oil / steel density	kg/m ³	963 / 7850	D
d_r / d_t	Rod / tubing ID	m	0.0222 (7/8") / 0.062 (2 7/8")	D

Conversions — 1 ksc = 98.07 kPa · 1 bbl = 0.159 m³ · 1 psi = 6.895 kPa · 1 cP = 0.001 Pa·s · $\nu(cSt) = \mu(cP)/SG$ · $SG = 141.5/(131.5 + ^\circ API)$

State vector for the twin: $x = [T_r, V_h, P_r, \mu_o, \lambda_o, q_{in}]$ with control input $u = [V_s, x, t_{soak}, N, S]$ and uncertainty σ . Every parameter carries value, unit, source, calibration method and confidence.

On the conflicting values

API gravity and native temperature differ between the problem statement and published field data. Preserve both rather than silently reconciling — but note that the field figures come from Oil India's own technical publications and SPE papers written by the field's operators, while the problem statement is an outreach document. Model with the field values; cite the discrepancy openly if asked.

Six Open items

1 · Second viscosity measurement — highest value

One measured point exists (11,500 cP at 50 °C). Fitting Walther requires two. $B = 3.5$ is assumed. The consequence is concentrated where it hurts: at 30 °C, $B = 3.2$ gives 96,600 cP and $B = 3.8$ gives 151,400 cP. The curves agree at steam temperature and diverge at reservoir temperature — precisely where the well spends most of its life.

2 · Calibrate the floating limit against failure history

Equation 5 is a static force balance. It ignores rod stretch, elastic wave effects, plunger fluid load and mechanical friction, so it reads optimistic — real $N_{\max}$ is lower. Oil India states it holds rod failure and pump unsettling history. Fit a correction factor so predicted $N_{\max}$ aligns with speeds at which failures actually occurred, and present it as calibrated rather than first-principles.

3 · Reservoir pressure has no governing dynamic

P_r appears in every state vector and drives q_{in} directly, but no dP_r/dt relationship is defined. Either add a simple depletion and support term, or state explicitly that it is held constant or externally supplied for v_l , with the reason.

Unmodelled, worth naming to judges: the crude is non-Newtonian — viscosity is quoted at a shear rate of 50 s^{-1} , so thickness depends on shear as well as temperature. Since rod motion through the fluid *is* a shear event, pump speed influences viscosity, which influences the safe pump speed. A feedback loop inside equation 5. Acknowledge as a known refinement; do not build it until the main chain works.